INCH-POUND

MIL-R-83726/25F(USAF)
9 August 1991
SUPERSEDING
MIL-R-83726/25E(USAF)
24 May 1985

MILITARY SPECIFICATION SHEET

(F) RELAYS, SOLID STATE, TIME DELAY (ON RELEASE) TRACK MOUNTED, TYPE IIA, CLASS C, SPST, 150 MILLIAMPERES, FIXED TIME, .050 TO 500 SECONDS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83726.

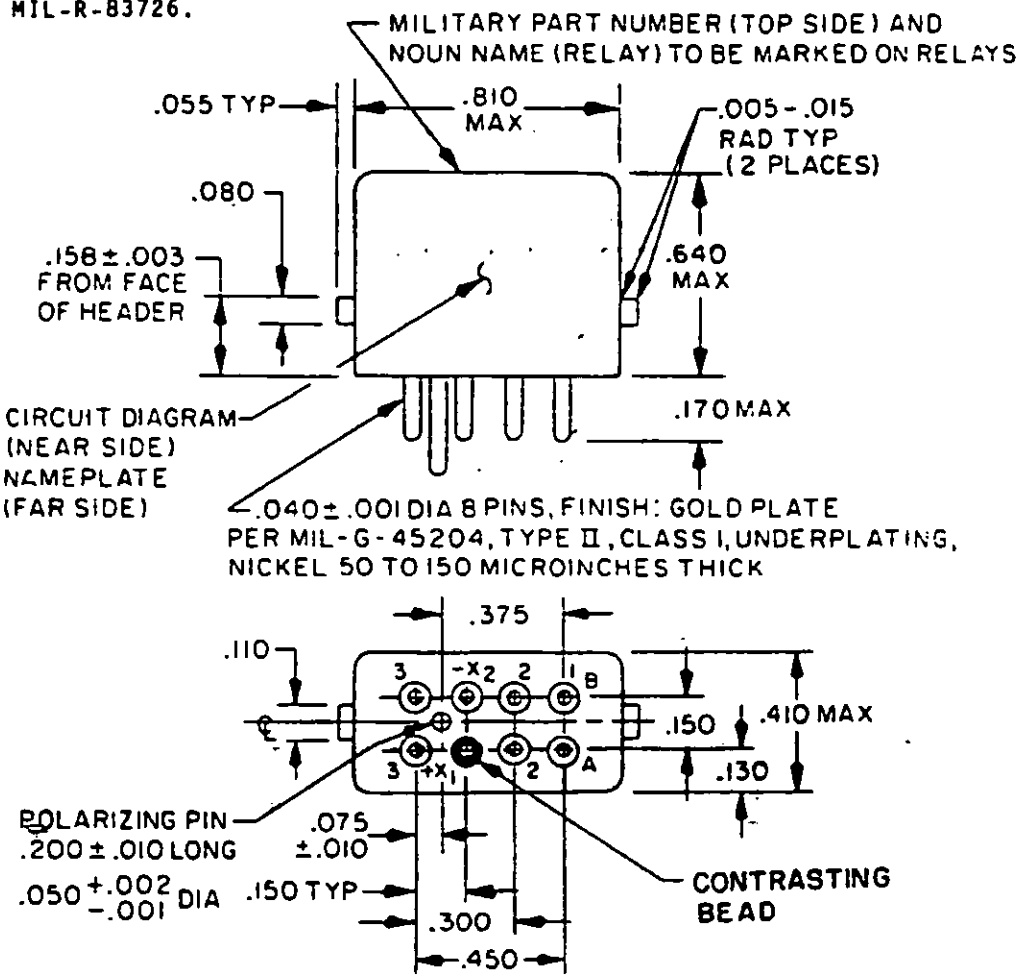


FIGURE 1. Outline dimensions and configuration.

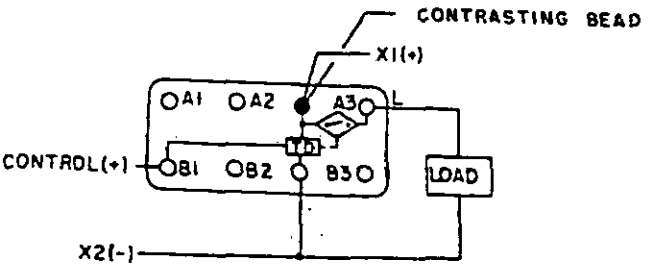
(F) denotes changes

FSC 5945

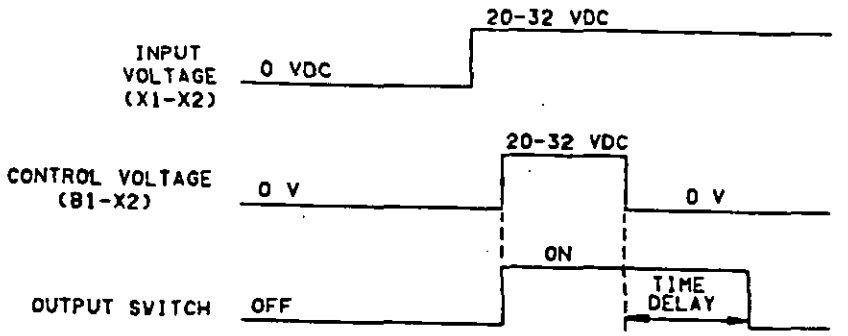
AMSC N/A

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CIRCUIT DIAGRAM
(SEE NOTES 4 AND 5)



TIMING DIAGRAM

Inches	mm	Inches	mm
.001	0.03	.080	2.03
.002	0.05	.110	2.79
.003	0.08	.130	3.30
.005	0.13	.150	3.81
.010	0.25	.158	4.01
.015	0.38	.170	4.32
.040	1.02	.200	5.08
.050	1.27	.410	10.41
.055	1.40	.640	16.26
.075	1.90	.810	20.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Terminal A1, A2, B1, and B2 are connected internally. Do not use for external tie points or for terminals.
5. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
6. Relays are intended for use with relay socket mount MIL-R-6106/23 and relay bracket mount MIL-R-6106/25.

FIGURE 1. Outline dimensions and configuration - Continued.

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REQUIREMENTS:

Operating data:

Timing action: Delay-on release.

Time delay: Fixed; select from .05 to 500 seconds.

Timing accuracy: ±10 percent of the nominal value. 1/

Recycle time: 10 milliseconds. 2/

Power interrupt: The accuracy will not be affected by power interruptions up to 1 millisecond spaced at least 10 milliseconds apart.

Control voltage (B1-X2): 20 V dc to 32 V dc.

Control current: 2 milliamperes maximum.

Input data:

Input voltage: 28 V dc nominal; range 20 V dc to 32 V dc.

Duty rating: Continuous.

Current drain: 10 milliamperes maximum plus load at 25°C.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

Output data:

Configuration: SPST; switch closure to +28 V dc.

Rating: 150 milliamperes maximum.

Suppression: Inductive suppression provided for output protection.

Voltage drop: 2 V dc maximum.

Leakage current: 1.0 microampere maximum at 28 V dc and 25°C; 10 microamperes maximum at 28 V dc and 125°C.

Endurance (life): 1,000,000 cycles minimum for test.

Qualification: 2,000 hours.

Group B: 1,000 hours or 1,000,000 cycles, whichever is less.

1/ The accuracy requirement applies for any combination of operating temperature and voltage.

2/ Recycle time is defined as the minimum time that power must be applied to the control terminal to assure that the next timing cycle will be completed within the specified timing tolerance. (Units can be recycled during timing or after time-out.)

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Electrical data:

Transients: 3/

Positive transients: +80 volts in accordance with MIL-STD-704, figure 9, limit 1.

Self-generated transients: ±50 volts maximum.

Spike susceptibility: ±600 volts, 10 microseconds maximum.

Electromagnetic interference: In accordance with MIL-STD-461, class 1D. 4/

Insulation resistance: 1,000 megohms at 500 V dc between each pin and case. 5/

Dielectric strength: 1,000 volts rms at 60 hertz at sea level between case and pins connected together. 5/

Environmental data:

Ambient temperature (operating and nonoperating): -55°C to +125°C.

Vibration (sinusoidal): 10 Hz to 55 Hz at 0.06 inch DA; 10 g's, 55 Hz to 500 Hz; 20 g's, 500 Hz to 3,000 Hz.

Ⓕ Vibration (random): MIL-STD-202, method 214, test condition 1B. Duration: 15 minutes each plane.

Shock: 1,100 g's for 0.5 millisecond, 1/2 sine, 3 axis.

Acceleration: 100 g's, 3 axes.

Seal: Hermetic.

Moisture resistance: MIL-STD-202, method 106.

Physical data:

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 0.56 ounce.

Ⓕ Marking: See MIL-R-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

3/ Transient specifications are based on a maximum duty cycle of 1/50.
4/ EMI test limits will not be exceeded during the timing interval or when continuously energized under steady-state conditions in accordance with Radio Interference requirements of MIL-R-83726.
5/ Terminals X1, X2 and B1 and A3 must be connected together during this test. Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and case.

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Ⓕ ESDS protection program: An ESDS protection program shall be implemented within 6 months of the date of revision F to this document. The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan which must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the EOS/ESD Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

Ⓕ ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.

Ⓕ ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

Part or Identifying Number (PIN): Consists of the prefix M83726/25-, a four-digit dash number (expressed in milliseconds), and a quality level indicator as follows:

M83726/25-0500W

Military designator: M

Specification sheet: 83726/25-

The first three digits are significant; the fourth digit is the number of zeros to follow the first three digits. The time is expressed in milliseconds and converted to seconds. (See examples.): 0500

Quality level indicator (W, X, or Y) 6/ (see MIL-R-83726): W

Examples: M83726/25-0500W - 50 millisecond time delay W level.
M83726/25-2503X - 250 second time delay X level.
M83726/25-5003Y - 500 second time delay Y level.

6/ Any relays numbered prior to the date of this specification without a quality level indicator shall be considered interchangeable (store and issue) with the "W" quality level.

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CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F713-08)

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